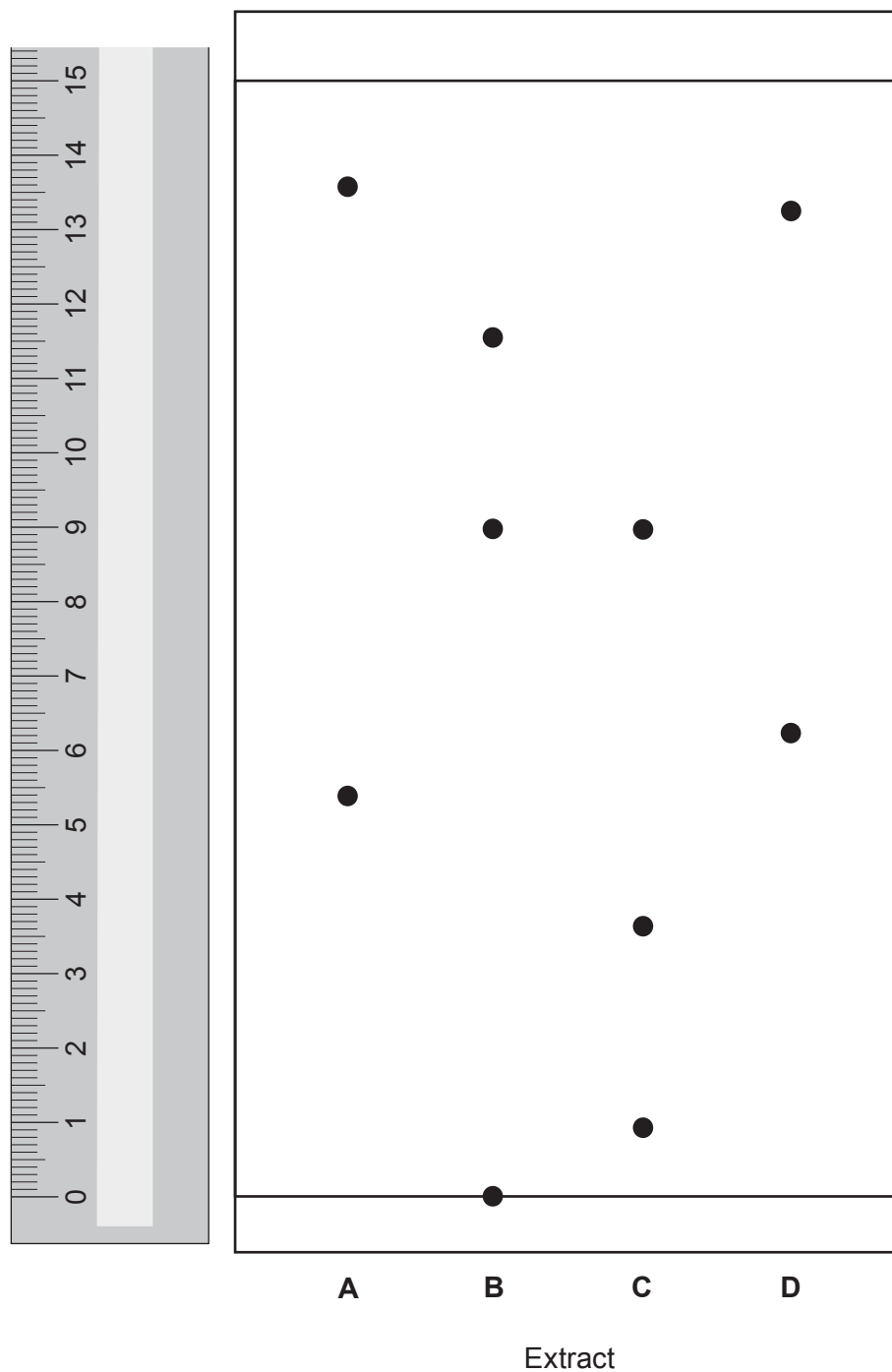


10. Paper chromatography can be used to identify plant leaf pigments. This process uses a chemical called acetone as the solvent instead of water.

The diagram shows the chromatogram of plant leaf extracts **A**, **B**, **C** and **D** in acetone.



- (a) All of the extracts contain a mixture of pigments with different R_f values.

For which plant leaf extract, **A**, **B**, **C** or **D**, is the **highest** R_f value 0.60?

Give your reasoning.

[3]

Extract

Reasoning

.....

.....

- (b) Explain why the pigments travel different distances on the chromatogram.

[2]

.....

.....

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- (c) One of the extracts contains a pigment which is insoluble in acetone.

State the **letter** of this extract. Explain your choice.

[2]

Extract

Explanation

.....

- (d) The chemical formula of the solvent acetone is C_3H_6O . Calculate the percentage by mass of carbon in acetone.

The relative formula mass (M_r) of acetone is 58.

[2]

$$A_r(C) = 12$$

Percentage = %

